2025-DNA-Methylation-Network

j2445wan

June 2025

1 Project Overview

This project aims to identify important genes methylations specific to 8 types of cancers, and build interaction network for these genes in each cancer type. Finally, networks will be compared to quantify differences in DNA methylations in cases with different cancers.

The project will be divided into 4 parts, and we are currently at stage 1 with a brief touch in stage 2.

1. Preprocess the data
2. Filtering out important genes (for each cancer types)
3. Build gene networks
4. Compare different networks

Github Repo Link: `https://github.com/Owen-Darkhorse/DNA-Methylation.git`

2 Data Source and Interpretation

The methylation data file name is Common_pan_cancer_hyper_bins_adjusted_and_normalized_cnt_in_SE.RDS and Common_pan_cancer_hyper_bins_adjusted_and_normalized_cnt_in_PE.RDS, respectively. From names, raw methylation counts have been bacth-corrected, DESeq2 normalized, and DMR screened by DESeq2 for hyper-methylated regions.

Each RDS file contains only 1 data matrix. Columns are indexed by sample IDs, and rows are indexed by genomic bin IDs. Note that each bin IDs not necessarily represent a single genes but a continuous stretch of DNA because the ID naming does not follow starndard Entrez or Ensembl gene IDs. These bin IDs are unique identifiers for fixed genomic regions (e.g., 300 bp windows) rather than gene identifiers. Data entries of methylation matrix are summarized methylation counts signal for that bin in that sample.

"The identification of DMRs in each comparison was limited in those 300 bp bins with more than 5 CpG sites, and required a fold change to be greater than 2 and false discovery rate (FDR) less than 0.05." Such a DMR identification was performed between batch correction and enrichment analysis. SE and PE samples were applied DMR analysis separately, 24418 hypermethylated DMRs overlapping SE and PE were identified as the pan-cancer signature. PCA and UMAP were used to visualize batch effect removal and sample normalization, instead of DMR selection.

Therefore, 'limma::lmFit' for DMR analysis has no need to be performed against to select differential genomic regions between healthy and cancer.

3 Literature Reviews

A pan-cancer compendium of 1074 plasma cell-free DNA methylomes and fragmentomes

1. Filters high quality data by saturation score ≥ 0.9, relH ≥ 3.0, and GoGe ≥ 3.0 in gene enrichment analysis
2. 6 methylation quantification and normalization methods were evaluated. ComBat-seq + DESeq2 was the most effective in batch effect removal. (The data we received)
3. Differentially methylated region analysis
   (a) Healthy vs. all cancers combined

(b) Healthy vs specific cancers

(c) Volcano plots for each cancer type

4. SE (single end) and PE (pair end) samples were analyzed separately, 24418 hypermethylated methylated regions (the same number of the features in the SE sample currently analyzed) were consistently detected in SE and PE samples.

5. Train cfDNA methylome and fragmentomic feature-based cancer classifiers

   (a) Calculated principal components (PCs) from methylation scores and 5 fragmenomic features (nucleosome footprinting, fragment ratios, end motifs, insert sizes)

   (b) Trained classifiers on extracted PCs derived from methylation and fragmentomic features.

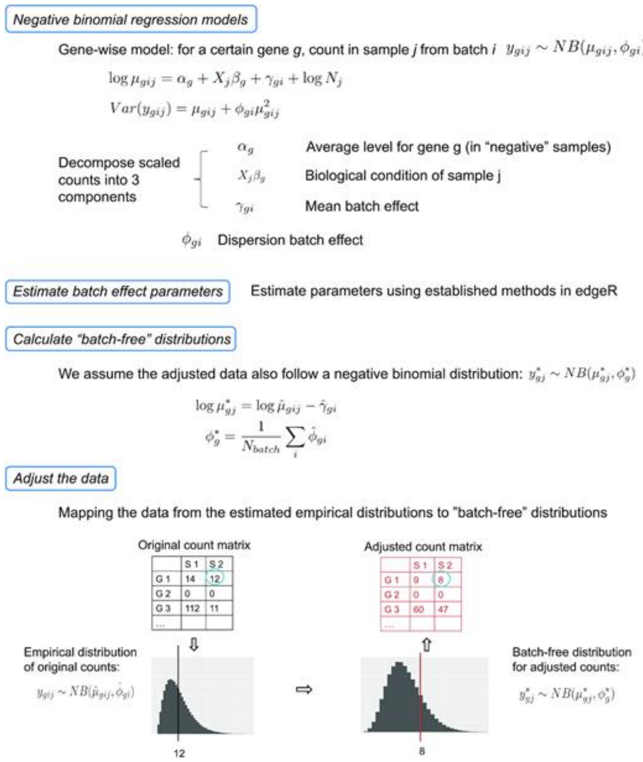

Figure 1: ComBat Procedures

regionalpcs improves discovery of DNA methylation associations with complex traits [?]

1. The goal: apply regional PCA as a better alterative to simple averaging to summarize methylation data more robustly and identify cell type-specific methylation changes associated with disease (such as Alzheimer's).

2. The regionalpcs method:

   (a) Divide methylations into groups based on their CpG locations: $X = \cup_{r=1}^{R} X_r$

   (b) IVT transform measurements of each methylation region

   (c) PCA on each X_r, obtain k components

   (d) Gavish-Donoho method to estimate the optimal number k of principal components needed to capture the essential variability in the data

   (e) Compute $Z_r = V_r X_r$, columns of Z_r are treated as rPCs

   (f) DM analysis

      i. Uses rPCs as the input matrix

      ii. The methylation was normalized using rank-based INT from the RankNorm function in the RNOmni package

      iii. 'limma::lmFit': identified differentially methylated genes

      iv. : 'limma::eBayes': obtain more stable estimates of the test statistics

v. Benjamini-Hochberg method for multiple test adjustments.
vi. lmFit model included age of death, PMI, study (ROS orMAP), batch, sex, and cell type proportions to construct the model matrix

3. Reference genome: Raw sequence reads were aligned to the GRCh37 human reference genome using the Burrows-Wheeler Aligner

Constructing a Cancer Patient-Specific Network Based on Second-Order Partial Correlations of Gene Expression and DNA Methylation

1. **Background**: epigenetic mutations are changes in DNA structure or genetic activities, but not in DNA sequence. Current sample-specific gene correlation network can only be constructed based on expression data, not epigenetic data.
2. **Research Problem**: how to identify genetic or epegenetic mutations specific to individual patients?
3. **Data Source:** TCGA: gene expression and methylation data from 10 types of cancers in solid normal tissues; Gene expression data were normalized using RNA-Seq by Expectation Maximization (RSEM) and converted to log2 scale
4. **Methods: Constructing the patient specific gene correlation network**
Let ρ_{xy} be the Pearson Correlation between expression levels of x and y. Let z, w be the third and the fourth variable under control, then
   - 1st order partial correlation between x and y, with z fixed
$$\rho_{xy.z} = \frac{\rho_{xy} - \rho_{xz}\rho_{yz}}{\sqrt{(1-\rho_{xz}^2)(1-\rho_{yz}^2)}}$$
   - 2nd order partial correlation between x and y, with z, w fixed
$$Pcorr_n(x,y|z,w) = \rho_{xy.zw} = \frac{\rho_{xw.z} - \rho_{xw.z}\rho_{yw.z}}{\sqrt{(1-\rho_{xw.z}^2)(1-\rho_{yw.z}^2)}}$$
where $\rho_{xw.z}$ is the first order partial correlation between x and w, with z fixed.
   - To test if the partial correlation is significant, use test statistics
$$T = \rho_{xy.zw}\sqrt{\frac{n-k-2}{1-\rho_{xy.zw}^2}}$$
where n is the number of samples, k is correlation order
   - Construct a reference network based on the the $Pcorr_n$ of gene expression AND methylation in n normal samples.
   - Add 1 tumor sample to the healthy sample to construct a perturbed network $Pcorr_{n+1}$, replace the tumor sample by another to construct the next
   - A partial correlation-based patient specific network (pcPSN) is the difference between the perturbed network and the reference network.
$$\Delta Pcorr = |Pcorr_{n+1} - Pcorr_n|$$
5. **Method: Finding Prognostic Gene Pairs From Patient Specific Networks**
   - Cox Proportional Hazard Model with $\Delta Pcorr$ of gene pairs
   - For each prognostic gene pair, cluster patients into 1) high $\Delta Pcorr$ groups: patients with $\Delta Pcorr \geq$ the average $\Delta Pcorr$ of the same gene pair of the same cancer type; 2) low $\Delta Pcorr$ group, and compare their survival rates.
6. **Topological and Functional Analyses of Patient-Specific Networks**
   - Extracted subnetworks from a pcPSN: composed of two genes of the gene pair and their neighbor genes
   - Annotation, Visualization and Integrated Discovery (DAVID) with the 100 most frequent genes in the subnetwork

- Clustering patient-Specific Networks of Various using Jaccard Index between graph edges:

$$D(N_i, N_j) = \frac{N_i \cap N_j}{N_i \cup N_j}$$

  Cancer Types

7. How do these methodology apply to ours
   - Instead of constructing patient specific networks, we can construct cancer-specific specific networks, with healthy controls set as the baseline.
   - For every pair of important genes, we can start with their first order partial correlation to construct the graph. WGCNA package is useful calculating an adjacency matrix from gene expression data using zero-order correlations
   - Instead of identifying co-methylations in all cancer types and healthy, we can just start with the healthy control and build cancer specific graphs upon there.

Analyzing integrated network of methylation and gene expression profiles in lung squamous cell carcinoma

- **Background**: The progressive accumulation of mutations and epigenetic abnormalities drive Lung squamous cell carcinoma progression. Only a few studies explored organizational and hierarchical interactions between these drivers.
- **Research Problem**: how to computationally reconstructed and integrated the diferentially expressed gene (DEG) network and the diferentially methylated cytosine (DMC) networks?
- **Data Source:** TCGA: Genomic Data Commons-Te Cancer Genome Atlas Lung Squamous Cell Carcinoma (GDC-TCGA LUSC) datasets
- **Method:** Network Characteristics
  1. Construct Differential expression analyses of gene expression (DEG) by the highest 1% of the partial information decomposition (PIDC), this yielded 7903 nodes and 3369 hypomethylated and 728 hypermethylated edges and cytosine methylation
  2. Using ELMER algorithm, DMC network represent the signifcant (adjusted-P ¡ 0.01) inverse correlation
  3. ELMER algorithm selects the closest 10 upstream genes and the closest 10 downstream genes for each DMC
- **Method: Community identifcation analysis**
  - Leiden algorithm identifies the 10 largest communities that had at least 200 nodes
  - Gene set enrichment analysis revealed the functional classes of each community
  - Strengths of interaction between the communities are measured by the ratio of the links connected between Communities x and y to the total number of intercommunity links on Community x, $C_x(y)$.
  - tested the signifcance of $C_x(y)$ using a network randomization test
- **Methods: Network Analysis**
  - Betweeness centrality (pivotal genes that controls flow of information) for each gene
  - Correlation between the number of genes that affect survival and the betweenness centrality
- **How can it apply to our research?**
  - Partial information decomposition calculate all possible edges between genes and their ranks
  - Betweeness centrality to identify the hub genes and characterize the network topology
  - How do occurrence of a specific cancer correlate with betweeness centrality of selected genes?
  - Enrichment analysis such as GSEA helps identify functional communities of gene methylations

Big Data, Small Bias: Harmonizing Diffusion MRI-Based Structural Connetomes to Mitigate Site-Related Bias in Data Integraion

- **What are structural connectomes?**
  Complex brain connectivity information is that is high dimensional and graph-structured and often follow a non-normal distribution.

- **Background:** Different scanners, acquisition protocols, preprocessing pipelines, and technical differences cause site-related variability in dMRI collected from different sites.

- **Proposal:** test several statistical harmonization frameworks to accomodate connectome data and evaluate their effectiveness in clearing site-related differences while preserving the connectivity information and biological association.

- **Method**
  - **Data**: connectome data from 6 datasets, 1194 typically developing children, 309 children with autism.
  - **Construct connectomes:** 86 regions of interest → 86 x 86 connectivity matrix.
  - **Harmonization Methods** Combat, log-Combat, Covbat, log-CovBat, gamma-GLM
  - **Evaluate Harmonization Methods:**
    - **Edgewise Site Effect Removal:** after regressing out sex and age effect from connectome strength, use Kruskal-Wallis test for detect edgewise group differences across sites, Brown-Forsythe test for variance difference between sites.
    - **On Removing Site Effects in Derived Graph Topological Measures:** node-level features (node strength, betweenness centrality, local efficiency, clustering coefficient) and global graph measures (global strength, intra- and inter-hemispheric strength, characteristic path length, global efficiency, modularity).

GRAPHICAL MODELS FOR ZERO-INFLATED SINGLE CELL GENE EXPRESSION

- **Authors:** Andrew McDavid, Raphael Gottardo, Noah Simon, Mathias Drton

- **Research Question:** how to infer co-regulatory networks from zero-inflated gene expression data.

- **Basic Idea:** genes can be "on" with positive expression reads, or it can be off with expression reads. We can model this as a finite mixture model of singular Gaussian distributions.
  The graph determining such a model describes each gene's statistical predictors: each gene is optimally predicted using only its neighbors in the graph.

- **Data Characteristics:** the empirical distribution of the log-transformed counts appears rather different than would be expected from censoring: the distribution of the log-transformed, positive values is generally symmetric. Yet the presence of bimodality in technically replicated experiments ("Pool/split" experiments) implicates the involvement of technical factors.

- **Method:** We propose a joint probability density function

$$\log f(\mathbf{y}) = \mathbf{v}_{\mathbf{y}}^T \mathbf{G} \mathbf{v}_{\mathbf{y}} + \mathbf{v}_{\mathbf{y}}^T \mathbf{H} \mathbf{y} - \frac{1}{2} \mathbf{y}^T \mathbf{K} \mathbf{y} - C(\mathbf{G}, \mathbf{H}, \mathbf{K}), \quad \mathbf{y} \in R^m$$

  -
$$\mathbf{Y} = [Y_1, \ldots, Y_m]$$
  -
$$\mathbf{V} = [V_1, \ldots, V_m]^T \equiv \left[I_{\{y_1 \neq 0\}}, \ldots, I_{\{y_m \neq 0\}}\right]^T$$
  - For any vector $\mathrm{v} = [v_1, \ldots, v_m] \, \varepsilon \{0, 1\}^{\mathrm{m}}$, define the subspace $R^{\mathrm{v}} = \prod_{i=1}^m R^{v_i}$ where we set $R^0 = \{0\}$. So, R^{v} is coordinate subspace corresponding to the non-zero entries of v.
  - $(\mathbf{Y} \mid \mathbf{V} = \mathbf{v}) \sim N(\mu(\mathbf{v}), \Sigma(\mathbf{v}))$ The normal distribution is singular
  - Joint probability of V (Ising Model): $p(\mathbf{v}) \equiv P(\mathbf{V} = \mathbf{v}) \propto \exp\left(\mathbf{v}^T \mathbf{G} \mathbf{v}\right), \quad \mathbf{v} \in \{0, 1\}^m$, where G is a symmetric interaction matrix in $R^{m \times m}$.
  - Joint distribution of Y given V: $\log f(\mathbf{y} \mid \mathbf{V} = \mathbf{v}) = \mathbf{v}^T \mathbf{H} \mathbf{y} - \frac{1}{2} \mathbf{y}^T \mathbf{K} \mathbf{y} - C'(\mathbf{H}, \mathbf{K}), \quad \mathbf{y} \in R^{\mathbf{V}}$
  - Marginal Distribution of Y: $f(\mathbf{y}) = \exp\{\mathbf{v}^T \mathbf{G} \mathbf{v} + \mathbf{v}^T \mathbf{H} \mathbf{y} - \frac{1}{2} \mathbf{y}^T \mathbf{K} \mathbf{y} - C(\mathbf{G}, \mathbf{H}, \mathbf{K})\}, \quad \mathbf{y} \in R^m$
  - In Ising model, the joint distribution of V has intractable normalization constant for large m, but the univariate full conditional distributions has tractable normalizing constants. So we let b be the interested coordinate and $A = \{1, ..., m\}/\{b\}$, $y_A = [y_i | i \in A]$
  - $\log f_{[b|A]}(\mathbf{y}) = v_b g_{[b|A]} + y_b h_{[b|A]} - \frac{1}{2} y_b^2 k_{[b|A]} - C_{[b|A]}, \quad y_b \in R$

– where $g_{[b|A]} = g_{bb} + 2\mathbf{g}_{bA}\mathbf{v}_A + \mathbf{h}_{bA}\mathbf{y}_A$

– $h_{[b|A]} = h_{bb} + \mathbf{h}_{Ab}^T\mathbf{v}_A - \mathbf{k}_{bA}\mathbf{y}_A$

– $k_{[b|A]} = k_{bb}$

– For any $a \in A$, define the parameter vector $_a = [g_{ba}, h_{ba}, h_{ba}, k_{ba}]$, By (12), $Y_b \perp Y_a | Y_{A/a}$ if and only if $\theta_a = 0$.

– Maximizing the penalized conditional log-likelihood function leads to a solution that is sparse in parameter blocks.

$$log f_{[b|A]}(y) - P_\lambda(\theta)$$

where the penalty is a group lasso penalty: $P_{\mathbf{H},\lambda}(\theta) = \lambda \sum_{a \in A} \sqrt{\theta_a^T \mathbf{H}_{aa} \theta_a}$. Here, $Hdiag(H_{aa})$ is a block-diagonal, positive-definite matrix that allows terms from the linear predictors to have different scales of penalty. It also accounts for correlation between components of $_a$, since columns of the design are correlated due to both v_a and y_a appearing as predictors.

4 Main Observations of Classical PCA

4.1 2D Representation

Figure 2 shows the scree plot, the scatter plot of score vectors, and trace plots of top 2 PCs of DNA methylation before any transformation. In the scree plot (A), most of variance were explained by the first few PCs. In the scatter plot (A), each point represents a subject, colors represent cancer types, coordinates of points represent score values of the subject's DNA methylation in the plane spanned by the top 2 PCs. Points seem to collapse into a funnel shape, and separation between cancers were not obvious. In C, D, line plots of the top 2 PCs, loading values exhibited spike patterns—with most of loading values near 0, a minority of genes gained absolute loading values that crosses 0.05 level. All PC 1 values are negative.

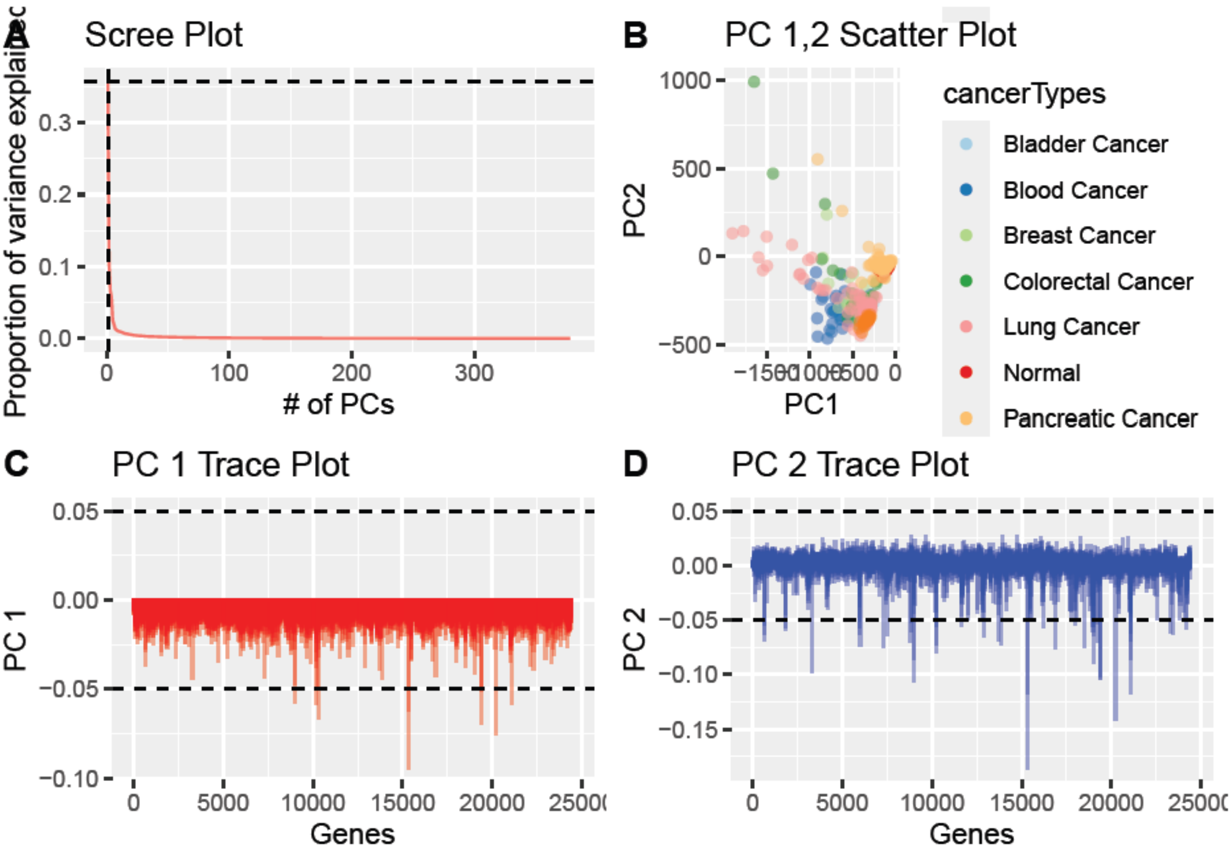

Figure 2: PCA 2D Graph of PCA before Any Transformation

After log transforming the raw methylation reads, blood cancer and lung cancer showed improved separation from the rest of cancer types. Points of pancreatic cancer are concentrated, but are overlapped with healthy normals. Breast cancer and colorectal cancer are mixed together.

All PC 1 values are negative, still. Spike patterns have been eliminated—loading values with larger absolute values tend to concentrate around the same level: 0.0125. PC 2 seem to be more volatile than PC 1, but loading values are generally stable.

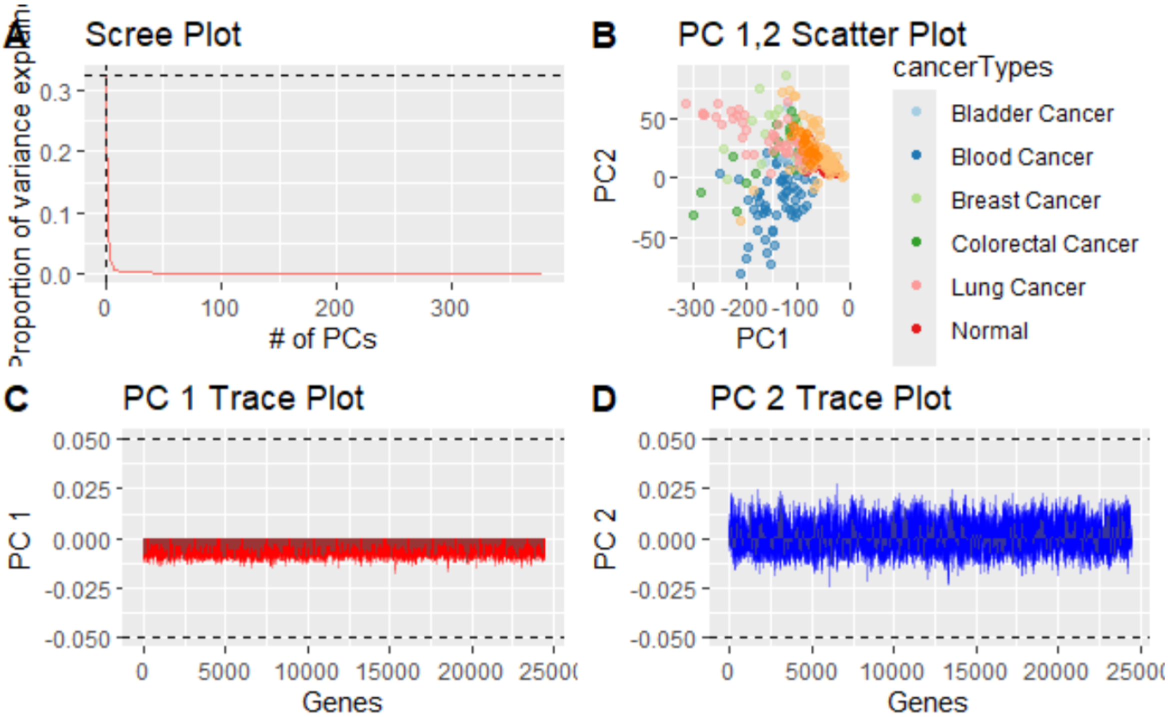

Figure 3: After Log Transformation

After applying IVT to every gene, points spread out from the initial funnel shape, and points of the same cancer type gather together more. In particular, blood cancer exhibited good seperations from the rest. Pancreatic cancer (yellow) gather in a clump, and showed overalap with healthy normal, breast cancer, and colorectal cancer. Renal cancers (in bright orange) gather in a small area and showed overlap with lung cancers (pink). The overlap could to perspective angle from which the projection is taken, 3D scatter plot will better help us view the class separation.

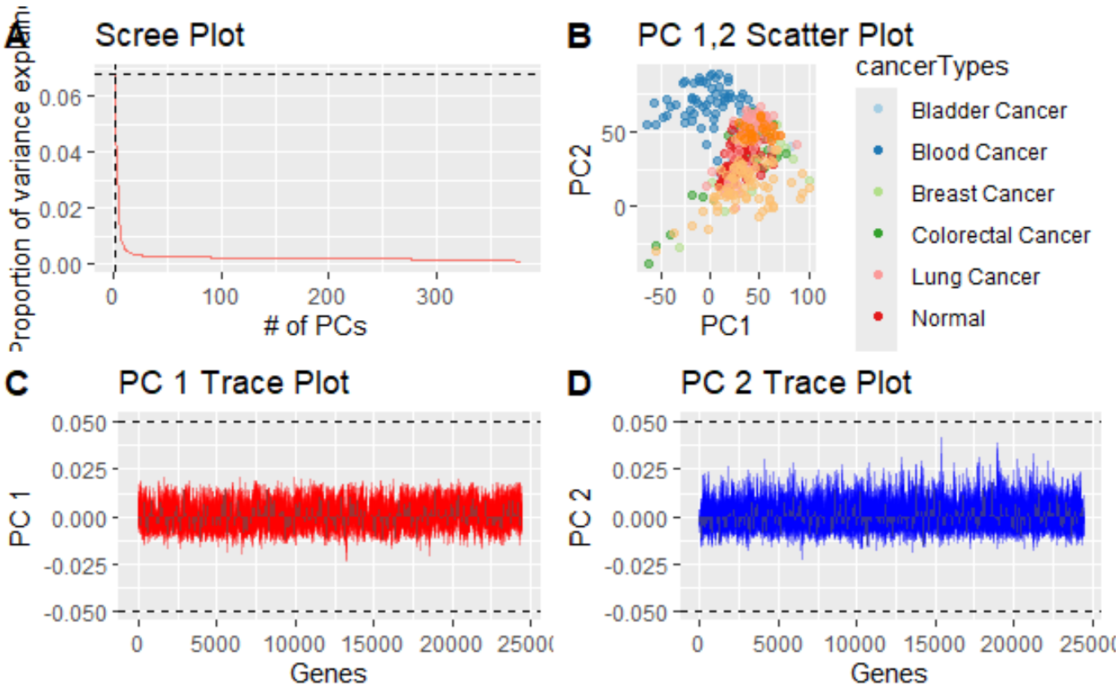

Figure 4: After IVT on Each Gene

4.2 3D Representation

Following plots graph distributions of healthy (blue) and cancer (green) before and after transformations. Rotation angles are different for inspect of details. In all 3 graphs, healthy subjects consistently locate at the corner of funnel-shaped distribution. Their locations are highly concentrated in the neiborhood. After log and IVT, points are more spread out, but they are restricted in near local as before transformation.

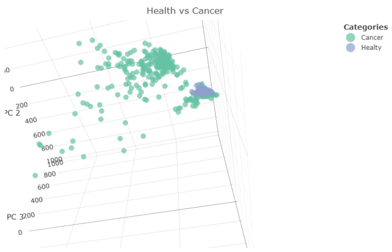

Figure 5: Before Transformation

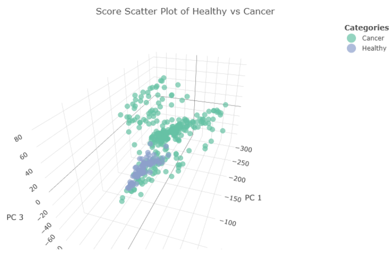

Figure 6: After Log

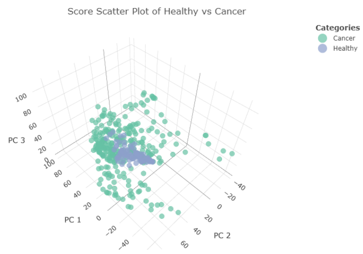

Figure 7: After IVT

Across 3 plots, normal and pancreatic cancers are crowded in the corner region of the funnel, while blood cancers are well separated from the rest. Lung cancers tend to locate on a hyperplane between blood cancers and normals, with renal cancers and breast cancers contained in between. Colorectal cancers are highly scattered between clouds of different cancer types.

After applying log and IVT, distributions of points are less funnel-shaped but more elliptical. After the log transformation, points showed a layered structure, with the bottom layer most be normal and pancreatic cancers, the middle layer be lung, breast, and renal cancers, and the top be blood cancer. After IVT, Healthy normal, pancreatic cancers, and blood cancers appear to locate on the same hyperplane, while lung cancer and breast cancers are on another plane.

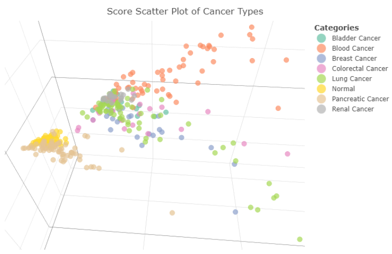

Figure 8: Before Transformation

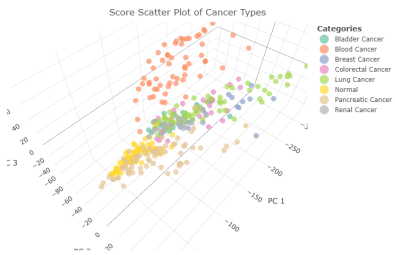

Figure 9: After Log

D-ivt.png

Figure 10: After IVT

5 Main Observations of Sparse PCA

5.1 2D Representation

We calculated the first 10 principal components after soft thresholding, and L1 norm of loading vectors are bounded by 10.

Before any transformation. The proportion of variance explained increases almost linearly with the number of PCs, such a trend plateaus at the 10th PC. Healthy normals, pancreatic, lung, renal, and colorectal cancers tend to cluster with points of the same labels, though overlaps between classes is extensive. PC 1,2 showed spike patterns with few genes' many values crossing -0.05 line. PC 1 is still all negative, while PC 2 has more negative values than positive. We might assume the separation boundaries between classes are non-linear.

After log, the proportion of variance explained by the top 10 PCs dropped from 40% before transformation to only 25%, the trend does not plateau at the 10th PC. In the scatter plot, points are collapsing into two clusters, with normals and pancreatic cancers in 1 cluster, and the rest in another cluster. Overlaps between classes are severe. PC 1 has only the negative part, PC2 has far more positive part than the negative part.

After IVT, the proportion of variance explained by the top 10 PCs dropped from 40% before transformation to only 25%. In the scatter plot, points are collapsing into a straight line, overlaps between classes are severe. PC 1 has only the negative part, PC2 has far more positive part than the negative part.

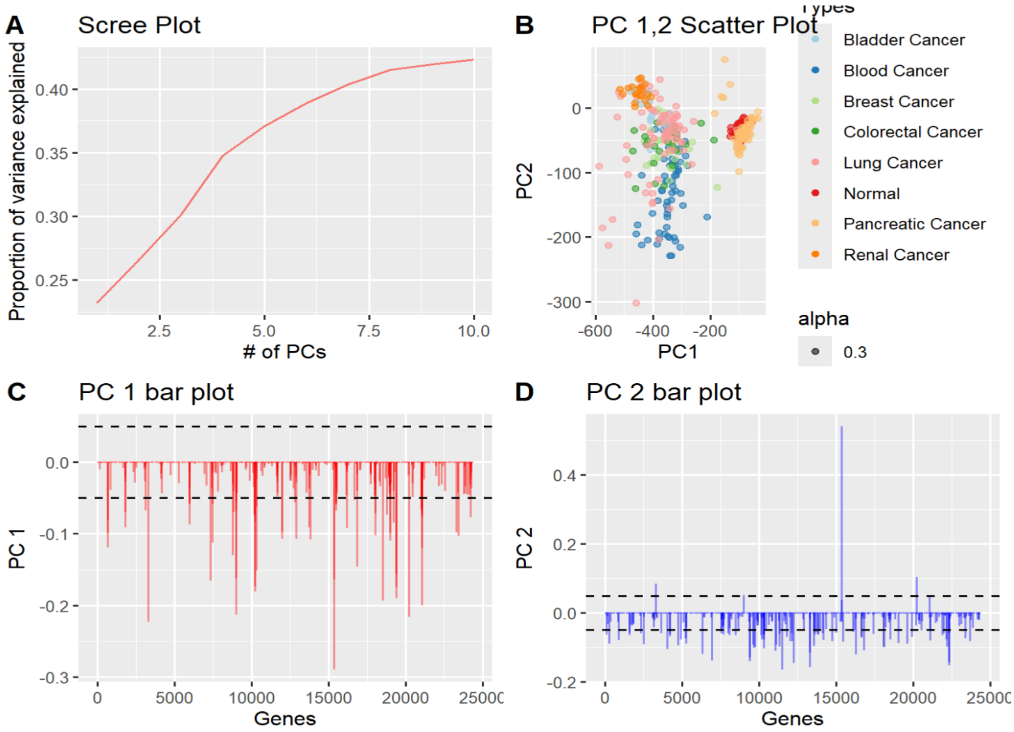

Figure 11: Sparse PCA before transformation

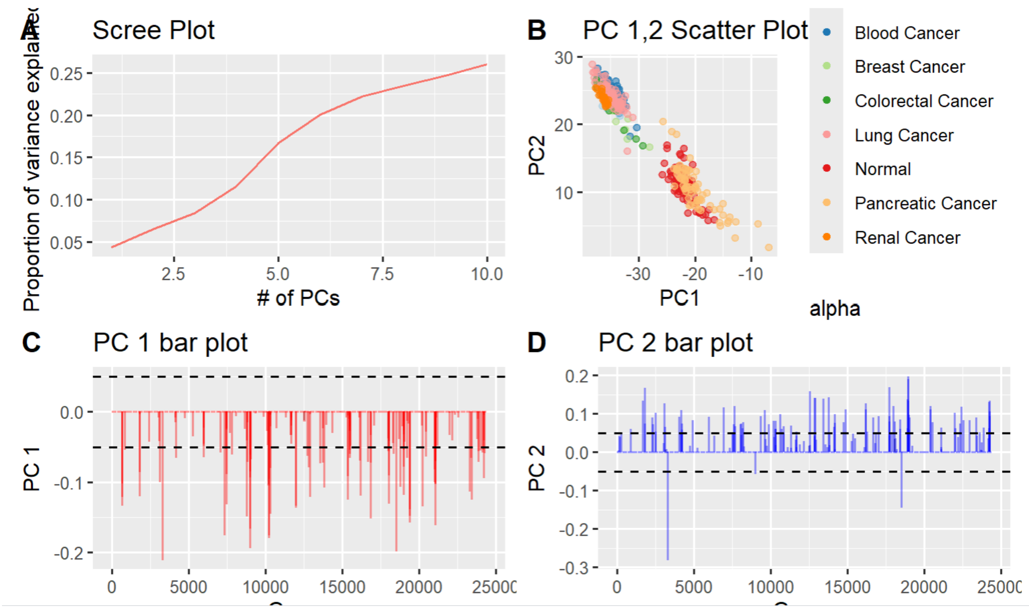

Figure 12: Sparse PCA after log

5.2 3D Representation

Points are scattered on a 2D plane before any transformation. They move toward a straight line after log or IVT is applied. The scale of score vectors changed from 0-400 to 0-40. Except clusters formed between normals, pancreatic cancers and the rest, classes move closer together, so the decision boundary become more circuitous.

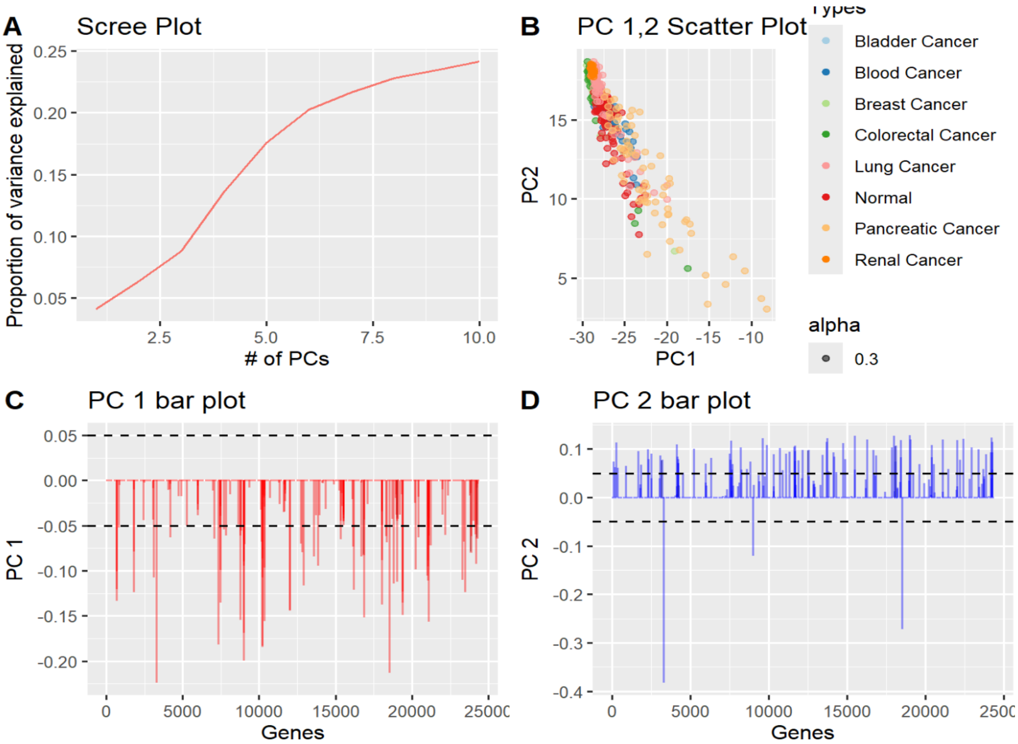

Figure 13: Sparse PCA after ivt

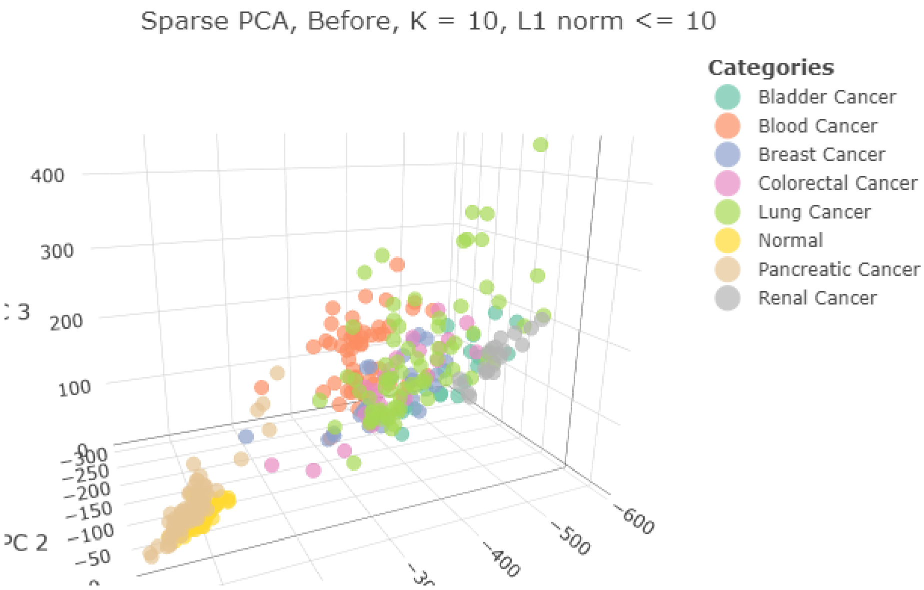

Figure 14: Sparse PCA before

6 Partial Correlation Graphs

6.1 Inverse Covariance-Derived Graph

6.2 Best Rho for the Null Graph

6.3 Visual Comparison of Graphs Computed from Different Cancer Types

6.4 Running Time Analysis

Elapsed time doesn't increase significantly with number of DMRs.

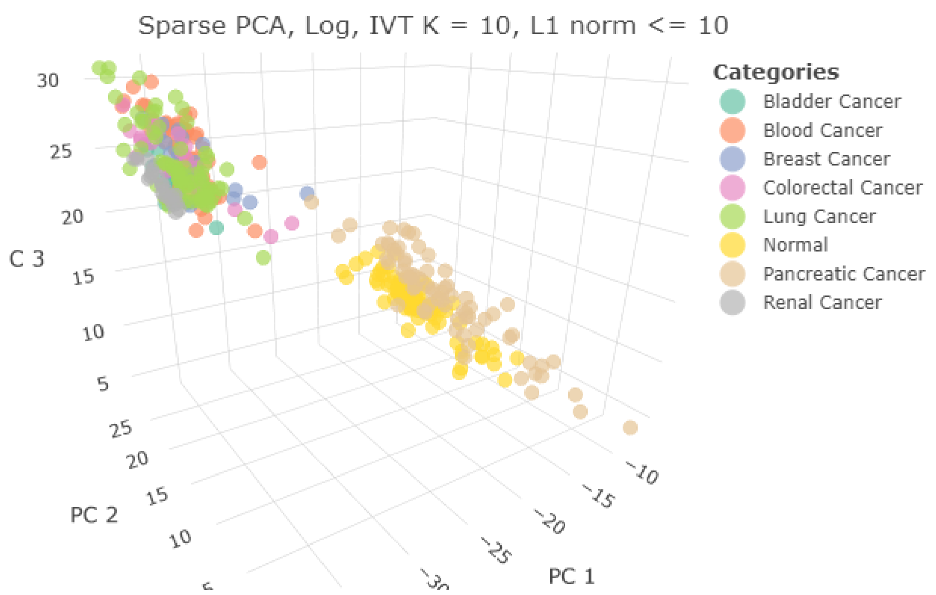

Figure 15: Sparse PCA after ivt

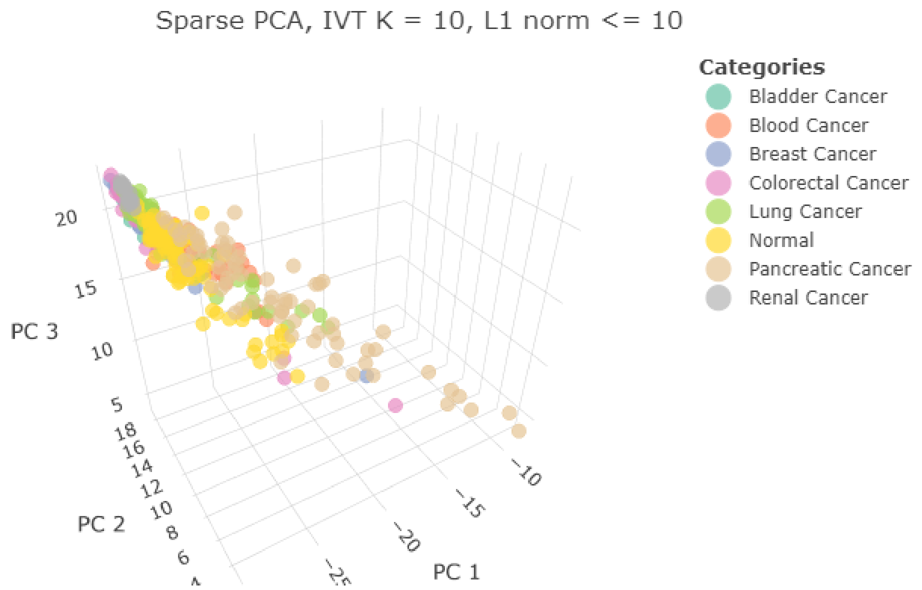

Figure 16: Sparse PCA after ivt

7 Addressing Niche Questions

7.1 Why PC1 of log Transformed Matrix is Negative?

The log transformation is $\log(x + 1)$. Plus 1 is a safeguard against the numerical error where x is 0 and NA is returned after log. After applying log, the distribution of methylation scores are still all positive right skewed. Since all methylations x's are scores are positive, $x + 1 \geq 1$, so $log(x + 1) \geq log(1) = 0$

After centering operation to every gene (column), the distribution becomes: Values are mixed signs. Does it imply that PC1's entries are mixed signs as well? For faster speed, I took the first 1000 genes, applied 'svd' to methylations of these genes, then I plotted distributions of PC1 and the first left singular vector. PC1 are all negative, whereas the left singular vector has mixed signs. This is counter-intuitive because PC1 typically captures the mean level of the data (near 0), and should be mixed-sign because the input matrix is mixed sign.

$d_1 u_1 v_1^T$ represents the overall level of the original matrix. Since the singular values are non-negative, sign of (i,j)-th entry in $d_1 u_1 v_1^T$ is controlled by u_1: positive if $u_{1i} < 0, v_{1j} < 0$. Since 36.77% of u_{1i} are negative, 36.77% values of $d_1 u_1 v_1^T$ are positive.

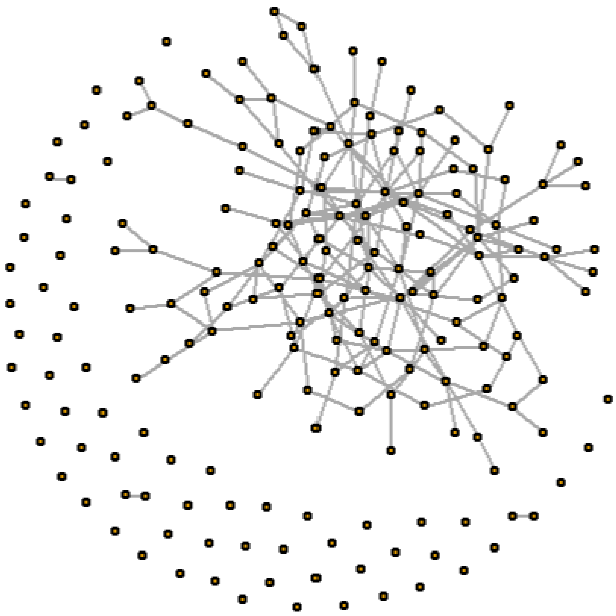

Figure 17: The graph derived from inverse covariance matrix on the top 200 genes in classical PCA. The inverse matrix was computed by glasso with penalty coefficient of 0.01. Edges correspond to matrix entries with the absolute value greater than 0.05. All 378 samples were used to compute this graph.

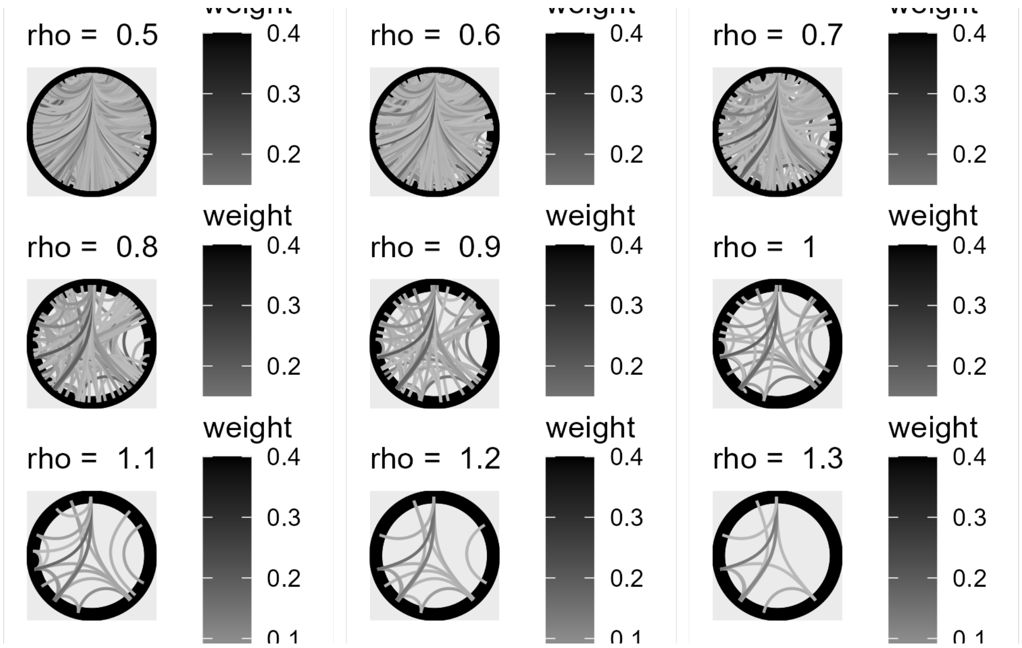

Figure 18: Null partial covariance graphs with rho 0.5 to 2. Colour of edges represent the partial correlation after penalty. Only healthy controls were modelled

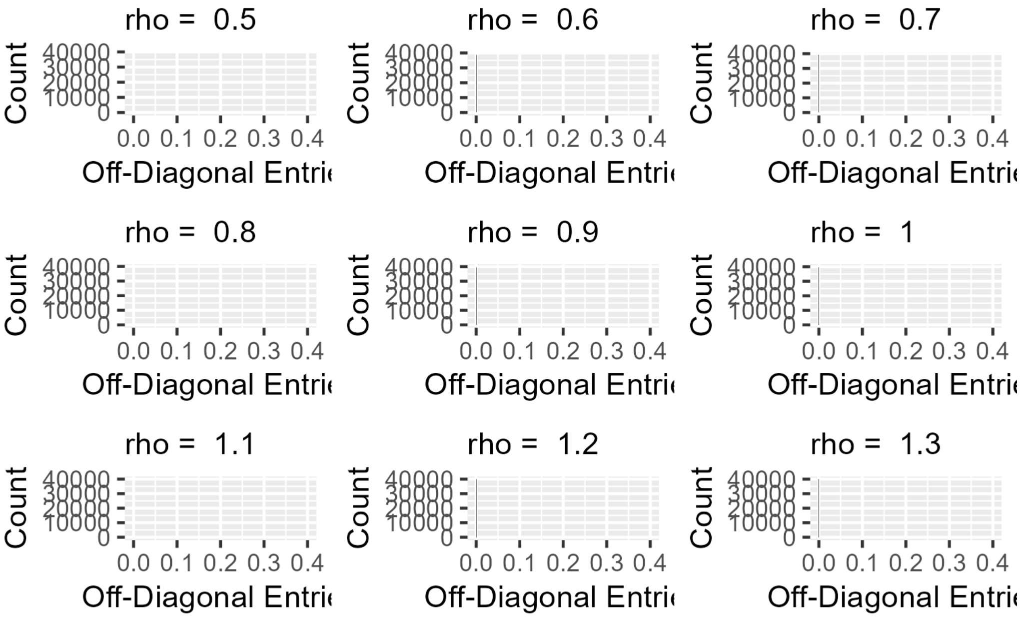

Figure 19: Value distribution of off-diagonal values in the partial covariance matrix with rho ranging between 0.5 to 2.

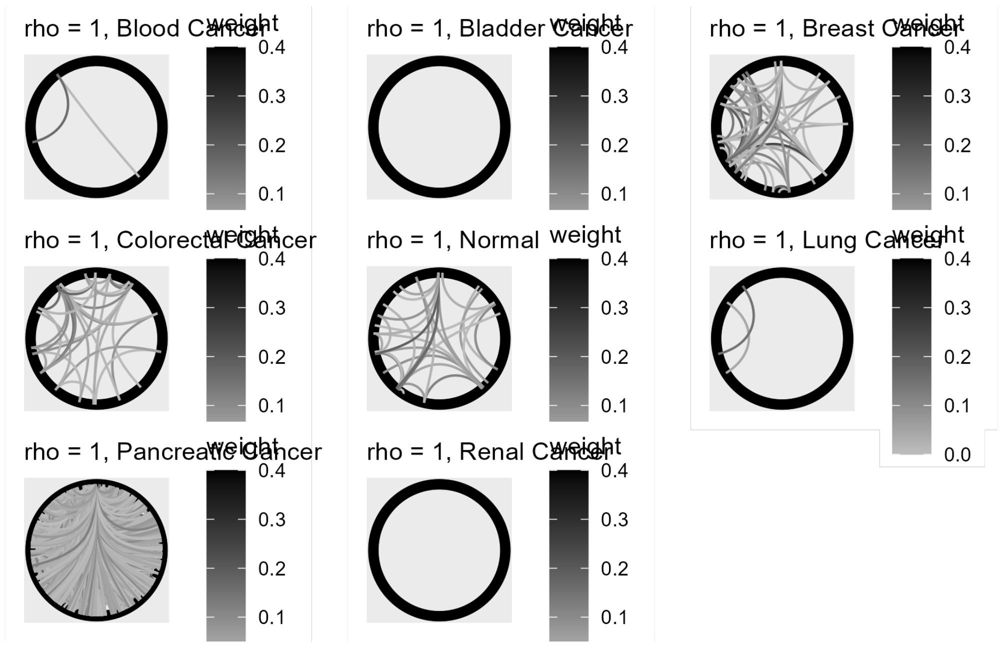

Figure 20: For $\rho = 1$ (optimal ρ in the null graph), graphs of the first 200 DMRs (sorted by B-statistics) were plotted. The line density of different cancer types are very different, which might imply that the optimal ρ need to be searched for different cancer types separately.

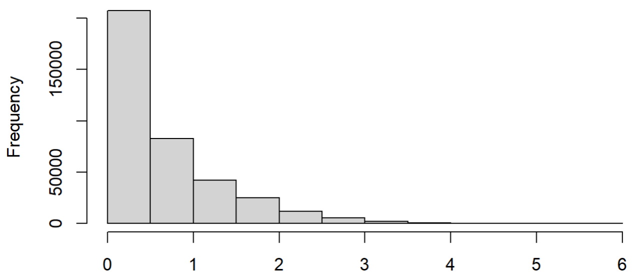

Figure 21: Right-skewed log Transformed Distribution

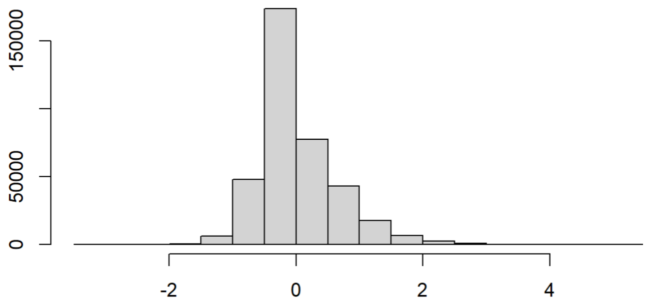

Figure 22: Column Centered log Distribution

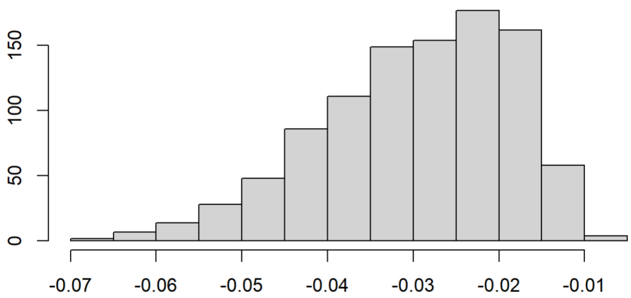

Figure 23: Distribution of PC1

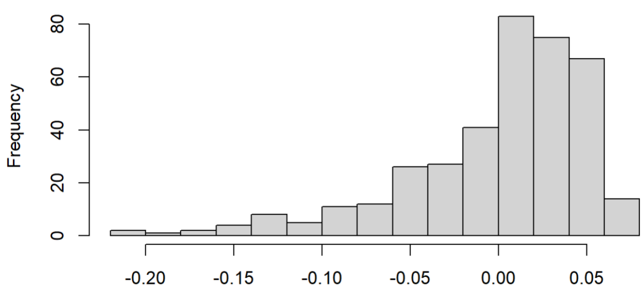

Figure 24: Distribution of the 1st Left Singular Vector

	Healthy		Cancer
p	Elapsed Time	p	Elapsed Time
200	6.1059	200	29.41579
400	6.3120	400	30.24477
600	5.4683	600	30.02641
800	6.6574	800	29.57831
1000	6.6716	1000	29.00682

Table 1: Elapsed Time of glasso for Healthy and Cancer groups with different number of DMRs included. The penalty coefficient $\rho = 1$

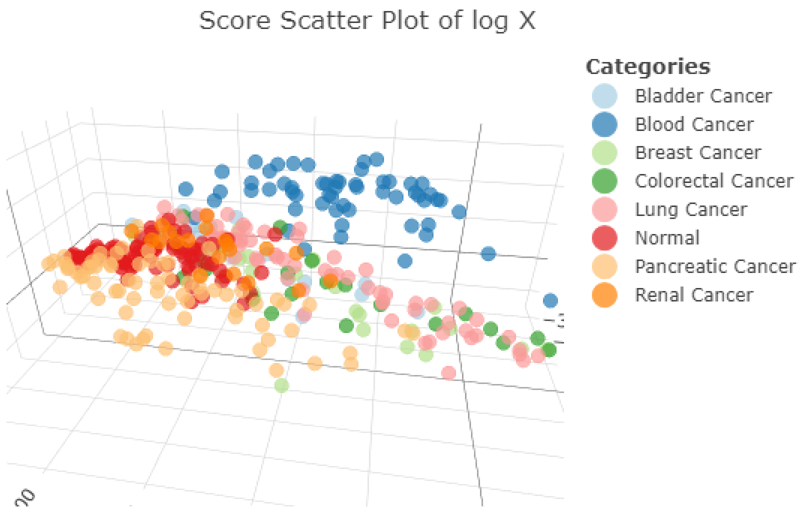

Figure 25: PCA 3D Scatter Plot After X is Centered before Score Matrix Computation

8 Data Pipeline

8.1 PCA and UMAP

8.2 Niche Question Investigation

This is the distribution close to 0 saw in the last time

2024-08-01 1. Filtering genes using B-statistics 2. Create the null network using healthy control only 3. Create the scenario where correlation of the selected genes are very close to 0.

2024-08-08 1. For different number of limma-selected genes, do computation power required in glasso step increase dramatically? 2. For each cancer type, take the difference between partial correlation of that cancer and healthy controls, plot the distribution of graph metrics (node strength, betweenness centrality, etc) to compare graphs. 3. If summary statistics act on weighted graphs, what variantion can we apply to partial correlation graphs?

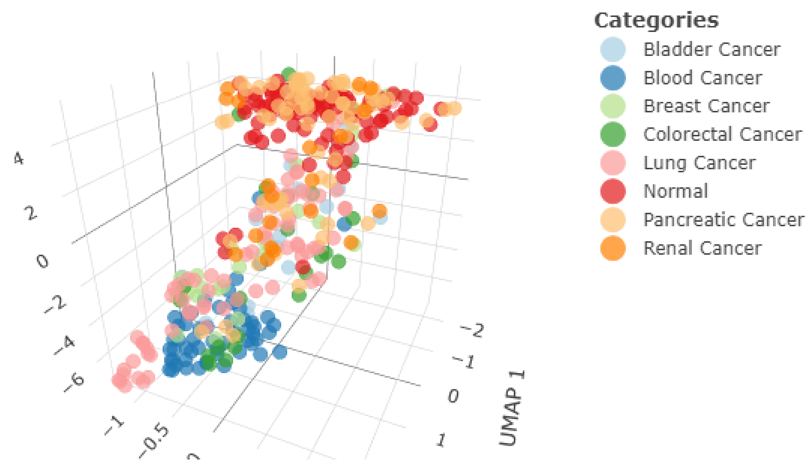

Figure 26: 3D UMAP Derived by the Top 200 Genes in PC1 Absolute Values

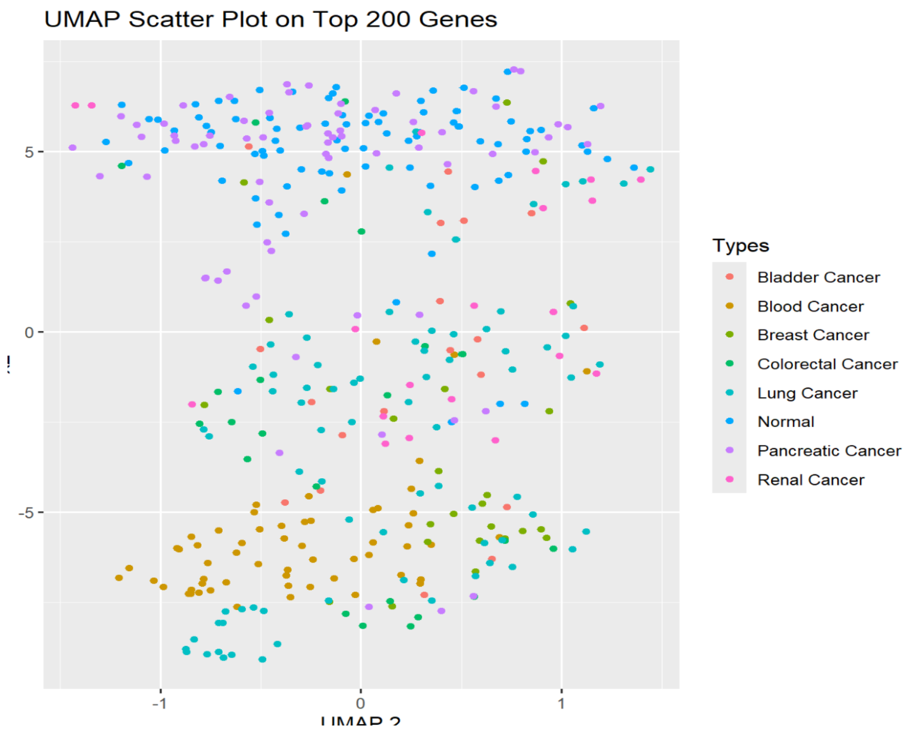

Figure 27: UMAP 2D Embedding of the Top 200 Genes

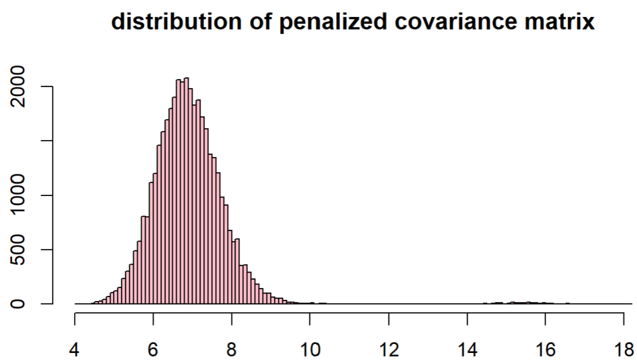

Figure 28: Distribution of glasso penalized covariance matrix between PCA-selected genes, for rho = 0.01

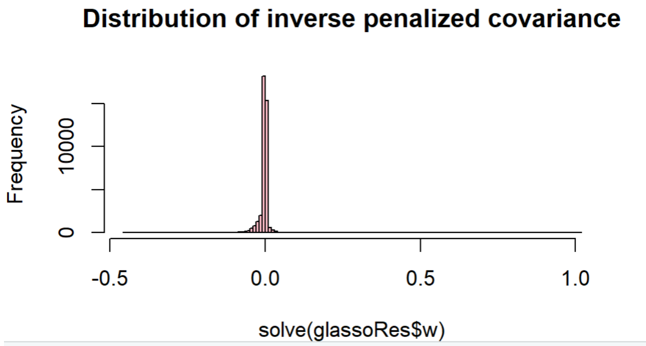

Figure 29: Distribution of inverse of glasso penalized covariance matrix between PCA-selected genes, for rho = 0.01. Most of values are close to 0.

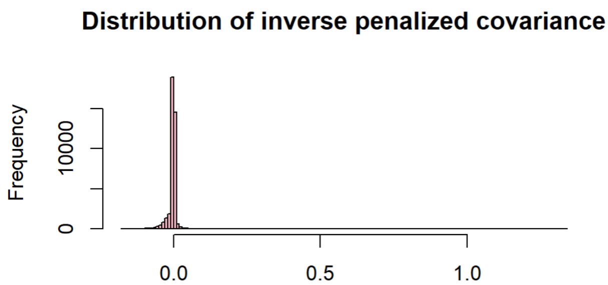

Figure 30: Distribution of inverse of glasso penalized covariance matrix between RANDOMLY selected 200 genes, for rho = 0.01. Most of values are close to 0.